

| Question | Answer                                                                                                               | Marks     |
|----------|----------------------------------------------------------------------------------------------------------------------|-----------|
| 4(a)     | <u>change</u> in internal energy = work done + energy transfer by heating                                            | <b>C1</b> |
|          | <u>increase</u> in internal energy = work done <u>on</u> system + energy transferred <u>to</u> the system by heating | <b>A1</b> |
| 4(b)(i)  | $U = (3 / 2) pV$                                                                                                     | <b>A1</b> |
| 4(b)(ii) | $pV = NkT$ <u>and</u> $k$ identified as Boltzmann constant                                                           | <b>B1</b> |
|          | $U = (3 / 2) NkT$                                                                                                    | <b>A1</b> |
| 4(c)(i)  | $W = (+)8XY$                                                                                                         | <b>A1</b> |
| 4(c)(ii) | $W = -20XY$                                                                                                          | <b>A1</b> |
| 4(d)     | work done during stages BC and DA = 0                                                                                | <b>B1</b> |
|          | change in internal energy (over complete cycle) = 0                                                                  | <b>C1</b> |
|          | thermal energy supplied = $20XY - 8XY$<br>$= (+)12XY$                                                                | <b>A1</b> |

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